

Assignment 1: AI-Augmented Business Briefing

Part A: Business Problem Identification

Name: Sai Nikhil Kannan

Student ID: 26235616

1. Dataset Overview and Business Context

The Diabetes Health Indicators Dataset contains health, lifestyle and socioeconomic data about individuals. There are several variables used in the dataset, that are grouped into different areas. Health related variables contain diabetes_binary, high blood pressure, high cholesterol, BMI, heart attack, and stroke. Lifestyle variables contain physical activity, fruit and vegetable intake, and heavy alcohol consumption. Socioeconomic variables contain age, gender, education and income.

From a business organisation perspective, the dataset offers valuable information for hospitals, NGOs, healthcare providers, and other health organisations. This dataset can be used by these business organisations for identifying people who are at higher risk of diabetes, recognise barriers to healthcare access and inform the screening programs. Organisations determine the groups that increased attention for diabetes related issues.

Overall, the dataset can support descriptive analysis of differences in diabetes status across health, lifestyle, demographic and healthcare-access characteristics. The findings could help organisations understand which groups or areas may require greater attention and support future decisions about screening, prevention and health-promotion programs. However, the dataset can identify associations between variables rather than establish that a particular factor directly causes diabetes.

Potential stakeholders who could use insights from this dataset are:

- Health insurance providers.
- Healthcare managers like a clinic or hospital.
- Public health policymakers.
- Fitness program providers

2. Potential Business Problems and Analytical Questions

Problem 1: Early Identification of People at Higher Risk of Diabetes

A healthcare organisation wants to identify population groups with characteristics associated with a higher likelihood of diabetes or pre-diabetes. Understanding these characteristics could help the organisation prioritise people for diabetes screening and health programs focused on prevention.

Why it matters:

Healthcare resources are limited, so identifying groups that may benefit from targeted screening or prevention programs could help organisations allocate resources more effectively.

Relevant features: Age, BMI, HighBP, HighChol, PhysActivity.

Decision supported: The analysis could help healthcare organisations decide which

population groups should be prioritised for diabetes screening, early intervention and prevention programs.

Analytical Questions:

1. How does diabetes status vary across different age groups?
2. How are BMI, high blood pressure and high cholesterol associated with diabetes status?
3. Do combinations of health characteristics, such as high BMI and high blood pressure, identify groups with a higher prevalence of diabetes?

Problem 2 : Evaluating Lifestyle Related Opportunities for Diabetes Prevention

Healthcare and public health organisations may want to understand whether lifestyle behaviours are associated with diabetes pre diabetes so they can design more focused health-promotion programs.

Why it matters:

The dataset contains information about physical activity, diet, smoking and heavy alcohol consumption. Understanding how these characteristics relate to diabetes status could help organisations decide which lifestyle areas should receive greater attention and advice in prevention programs.

Relevant features: PhysActivity, Fruits, Veggies, Smoker, HvyAlcoholConsump, BMI, HighBP, HighChol, GenHlth.

Decision supported: The analysis could help healthcare and public health organisations decide which lifestyle areas and population groups should be prioritised in diabetes prevention and health-promotion programs.

Analytical Questions:

1. How does diabetes status differ between people who report regular physical activity and those who do not?
2. How is diabetes status associated with reported fruit and vegetable consumption?
3. Do people with multiple lifestyle-related characteristics, such as physical inactivity and smoking, show a higher prevalence of diabetes?

Problem 3: Understanding Healthcare Access and Its Relationship with Diabetes Outcomes

Healthcare organisations and policymakers may need to understand whether barriers to accessing healthcare are associated with diabetes and pre diabetes status.

Why it matters:

The dataset includes variables relating to healthcare accessibility and affordability. If certain groups experience barriers to healthcare access and also show different diabetes outcomes, organisations could use this information when designing screening or outreach programs for them.

Relevant features: AnyHealthcare, NoDocbcCost, CholCheck, Age, Income, Education, GenHlth, HighBP, HighChol.

Decision supported: The analysis could help healthcare organisations and policymakers decide which groups may require targeted healthcare outreach, screening or support to reduce potential barriers to diabetes care.

Analytical Questions:

1. How does diabetes status differ between people with and without healthcare coverage?
2. Is the inability to visit a doctor because of cost associated with different diabetes outcomes?
3. Do healthcare-access indicators vary across income and education groups, and how could these differences relate to diabetes screening needs?

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Part B: AI Collaboration Reflection

Name: Sai Nikhil Kannan

Student ID: 26235616

AI Collaboration Reflection:

I used ChatGPT for my assessment. Firstly, I uploaded the assessment requirements and dataset file to understand the assessment thoroughly and how to proceed further with it. ChatGPT suggested three problems to me, and I used the second and third problems.

I submitted my first problem to ChatGPT myself to check whether it aligns with the dataset and took the suggestions.

I found it hard to link the variables to the right business questions. I also wasn't sure if I had to use the exact variable names from the dataset when forming my business problems and analytical questions. So, I asked ChatGPT for ideas on possible business problems, analytical questions, and relevant variables. I sought advice on how to present the variables correctly.

Prompts used:

Assessment Task 1: AI-Augmented ...
PDF

Data Description(In)(1)

	Feature Name	Allowed / Value Rang	Description	
1	Diabetes_binary	0 = No diabetes (or pregnancy only) 1 = Prediabetes or Diabetes	Target variable indicating diabetes status	
2	HighBP	0 = No 1 = Yes	Ever told by a doctor that you have high blood pressure	
3	HighChol	0 = No 1 = Yes	Ever told by a doctor that you have high cholesterol	

This is my assessment task 1. I don't have to run a full analysis, and I am allowed to use GenAI Tools. Given that data description csv file of dataset.

Yes. I've reviewed the **assessment instructions** and the **data-description CSV**. You are correct: **you are not expected to perform a full statistical or machine-learning analysis**. The task is mainly about showing that you can look at the available variables, identify realistic organisational problems, and formulate analytical questions that could be investigated later.

The assessment specifically asks for **three potential business problems**, with each linked to:

- why the problem matters,

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Problem 1: Early Identification of People at Higher Risk of Diabetes

Problem statement: A healthcare organisation wants to find people who may be more likely to have pre-diabetes or diabetes. This way, these individuals can be prioritised for screening and health programs focused on prevention

Worked for 6s >

Yes — **this is a strong direction for Problem 1**. I would make one small change: since your dataset contains a diabetes diagnosis indicator, but you are **not conducting analysis**, avoid saying the organisation can actually "find" people who are at higher risk. Frame it as identifying **characteristics associated with higher diabetes status** that could later support screening prioritisation.

What AI Suggestions Were Useful

The AI suggestions helped identify relevant variables for each business problem and improved the link between the dataset and possible healthcare decisions. For example, the suggestion to treat Diabetes_binary as the outcome variable instead of an explanatory feature made the analytical problem clearer. The AI also helped me separate lifestyle, health, demographic, and socioeconomic variables.

What Suggestions Were Rejected

I did not accept all AI suggestions without thinking. For instance, I rejected the idea of describing the dataset as directly identifying people with pre-diabetes because the Diabetes_binary variable does not specify a pre-diabetes category. I also avoided claiming that lifestyle factors cause diabetes since the proposed analysis focuses on identifying associations rather than proving causation.

How the AI Output Was Checked or Improved

I checked the AI suggestions against the dataset description and the actual variables in the dataset. I also thought about whether each variable logically connected to the specified business problem. I then revised the wording to ensure that the business problems were realistic and that the claims aligned with what the dataset could support.

How My Thinking Changed After Using AI

Using AI helped me think beyond just listing variables in the dataset. I started considering how the variables could inform real organisational decisions, such as prioritising groups for diabetes screening or pinpointing areas for prevention programs. It also helped me see the importance of distinguishing between an outcome variable and explanatory variables when creating an analytical business problem.